22IT301/22CS301/22IS301/22CT301/22CD301 - PROBABILITY, STATISTICS AND QUEUING THEORY

UNIT II – STANDARD DISTRIBUTIONS

2.6 NORMAL DISTRIBUTIONS

2.6.1 INTRODUCTION

The Normal Distribution was introduced by the French Mathematician Abhraham De Moivre in 1733. De Moivre, who used this distribution to approximate probabilities connected with coin tossing, called if the exponential bell-shaped curve.

DEFINITION: A Normal distribution is a continuous distribution given by

$$y=\frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

where X is a continuous normal variate distributed with density function $f(x)=\frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$ with mean μ and standard deviation σ

2.6.2 STANDARD NORMAL PROBABILITY DISTRIBUTION:

If X is a normal distributed random variable μ and σ are respectively its mean and standard deviation, then $Z=\frac{X-\mu}{\sigma}$ is called standard normal random variable.

2.6.3 MEAN AND VARIANCE OF STANDARD NORMAL DISTRIBUTION:

- **Mean** $E[X]=\mu=0$
- **Mean Square** $E[X^2]=1$
- **Variance** $Var[X]=1$

2.6.4 PROBLEMS BASED ON NORMAL ISTRIBUTION

Normal Distribution :

The normal distribution is the most important probability in Statistics since it fits many natural phenomena like heights, blood pressure, measurement error and IQ scores follow the normal distribution. It is also known as the Gaussian distribution and the bell curve.

The normal distribution is a probability function that describes how the values of a variable are distributed. It is a symmetric distribution where most of the observations cluster around the central peak and the probabilities for values further away from the mean taper off equally in both directions. Extreme values in both tails of the distribution are similarly unlikely. Normal distribution also the limiting form of Binomial distribution.

Definition :

A continuous random variable x with probability density function

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}, \quad -\infty < x < \infty, \quad -\infty < \mu < \infty, \quad \sigma > 0$$

is said to follow normal distribution.

Here the parameters are σ and μ, where μ = mean & σ = S.D.

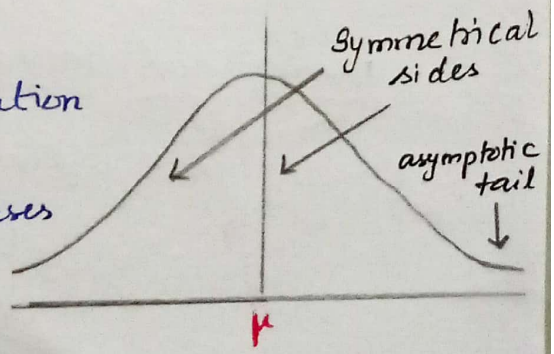
The normal distribution is denoted by $X \sim N(\mu, \sigma^2)$.

Characteristics of Normal distribution and Normal probability curve

(i) The curve is bell shaped and symmetrical about the line $x = \mu$ mean = median = mode

(ii) Mean, Median and Mode of the distribution Coincide.

(iii) As x increases numerically, $f(x)$ decreases rapidly the maximum probability occuring at the point $x = \mu$, given by $[p(x)]_{max} = \frac{1}{\sigma\sqrt{2\pi}}$.



(iv) Since $f(x)$ is non-negative, the curve will not go below the x-axis.

(v) x-axis is asymptote to the curve.

Moment Generating function of Normal distribution:

By definition, $M_X(t) = E[e^{tx}]$

$$= \int_{-\infty}^{\infty} e^{tx} f(x)\, dx = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{tx} \cdot e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx$$

put $z = \frac{x-\mu}{\sigma} \Rightarrow x = \mu + \sigma z \Rightarrow dx = \sigma\, dz.$

$$\therefore \quad M_X(t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{t(\mu+\sigma z)} \cdot e^{-z^2/2}\, dz$$

$$= e^{\mu t} \cdot \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(z^2 - 2t\sigma z)}\, dz.$$

$$= e^{\mu t} \cdot \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}\left[(z-\sigma t)^2 - \sigma^2 t^2\right]}\, dz.$$

$$= e^{\left(\mu t + \frac{\sigma^2 t^2}{2}\right)} \cdot \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(z-\sigma t)^2}\, dz.$$

put $u = z - \sigma t \Rightarrow du = dz$

$$\therefore \quad M_X(t) = e^{\left(\mu t + \frac{\sigma^2 t^2}{2}\right)} \cdot \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{u^2}{2}}\, du$$

$$= e^{\left(\mu t + \frac{t^2\sigma^2}{2}\right)} \cdot \frac{2}{\sqrt{2\pi}} \int_{0}^{\infty} e^{-\frac{u^2}{2}}\, du.$$

put $t^2 = \frac{u^2}{2} \Rightarrow u = t\sqrt{2} \Rightarrow du = dt\sqrt{2}.$

$$\therefore \quad M_X(t) = e^{\left(\mu t + \frac{t^2\sigma^2}{2}\right)} \cdot \frac{2}{\sqrt{2\pi}} \int_{0}^{\infty} e^{-t^2}\, dt\sqrt{2}$$

$$= e^{\left(\mu t + \frac{t^2\sigma^2}{2}\right)} \frac{2}{\sqrt{\pi}} \int_{0}^{\infty} e^{-t^2}\, dt = e^{\left(\mu t + \frac{t^2\sigma^2}{2}\right)} \cdot \frac{2}{\sqrt{\pi}} \cdot \frac{\sqrt{\pi}}{2}$$

$$\therefore \quad M_X(t) = e^{\mu t + \frac{t^2\sigma^2}{2}}$$

Mean and variance of Normal distribution

$$M_X(t) = e^{\mu t + \frac{\sigma^2 t^2}{2}}$$

$$E(X) = \left[\frac{d}{dt} M_X(t)\right]_{t=0} = \left[e^{\mu t + \frac{\sigma^2 t^2}{2}} (\mu + \sigma^2 t)\right]_{t=0}$$

$$= e^0(\mu + 0) = \mu.$$

$$\boxed{\text{Mean} = \mu}$$

$$E(x^2) = \left[\frac{d^2}{dt^2} M_x(t)\right]_{t=0} = \frac{d}{dt}\left[e^{\mu t + \frac{\sigma^2 t^2}{2}}(\mu + \sigma^2 t)\right]_{t=0}$$

$$= \left[e^0(\mu)^2 + e^0\sigma^2\right]$$

$$E(x^2) = \mu^2 + \sigma^2$$

$$\therefore \text{Variance}(x) = E(x^2) - (E(x))^2 = \mu^2 + \sigma^2 - \mu^2 = \sigma^2.$$

$$\therefore \boxed{\text{Variance}(x) = \sigma^2.}$$

Median of normal distribution :

M is the median of x if $\int_{-\infty}^{M} f(x)\,dx = \int_{M}^{\infty} f(x)\,dx = \frac{1}{2}$.

Here $\boxed{\text{median} = \mu.}$

Mode of Normal distribution :

Mode of a continuous random variable is defined as the value of x for which $f(x)$ is maximum.

i.e, $f''(x) < 0$; $f'(x) = 0$.

Here $f(x)$ is maximum at $x = \mu$.

$$\therefore \boxed{\text{Mode} = \mu.}$$

Standard Normal Distribution

If X is a random variable following normal distribution with parameter μ and σ, then $z = \frac{x-\mu}{\sigma}$ is called the standard normal variate and the pdf of the standard variate z is given by

$$\phi(z) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{z^2}{2}} \quad ; \quad -\infty < x < \infty.$$

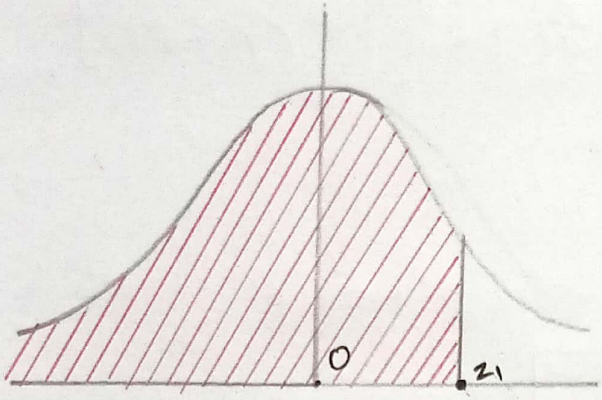
Properties of Standard Normal curve :

(i) Total area under the standard normal curve is equal to 1.

(ii) The standard normal curve is symmetric about 0.

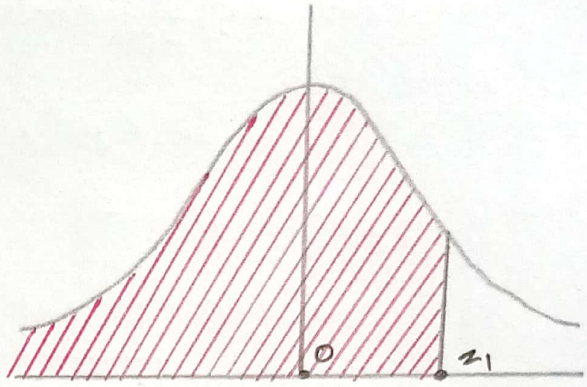
(iii) Most of the area under the standard normal curve lies b/w −3 & 3.

z	Area under curve between −z and z	% of total area
1	0.6826	68.26 %
2	0.9544	95.44 %
3	0.9974	99.74 %



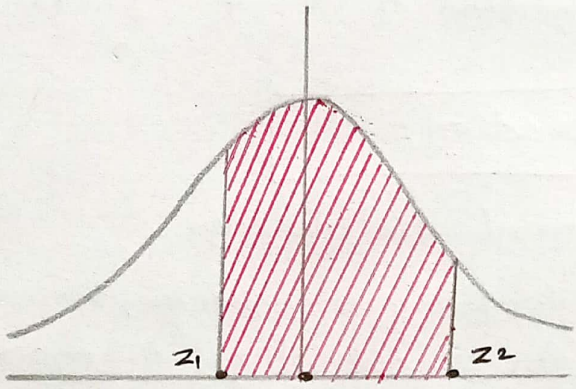
Standard area $= P(z < z_1)$

$= 0.5 + P(0 < z < z_1)$.



Standard area $= P(z > z_1)$

$= 0.5 - P(0 < z < z_1)$.



Standard area $= P(z_1 < z < z_2)$

$= P(0 < z < z_1) + P(0 < z < z_2)$.

Problems:

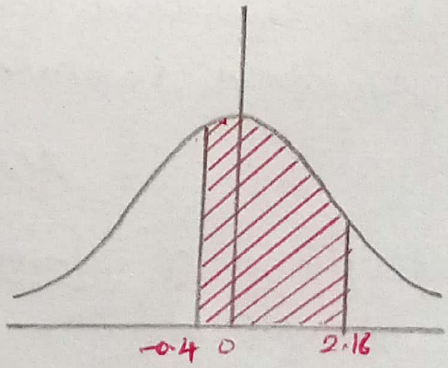
1. Let x denote the number of grams of hydro carbons emitted by an automobile per mile. Assuming x to normal with $\mu = 1$ gram and $\sigma = 0.25$, find the probability that a randomly selected automobile will emit between 0.9 and 1.54 gram of hydro carbons per mile.

Solution: Given: $\mu = 1$ & $\sigma = 0.25$

To find $P(0.9 \le x \le 1.54)$.

Let $z = \dfrac{x-\mu}{\sigma} = \dfrac{x-1}{0.25}$

$$P(0.9 \le x \le 1.54) = P\left[\frac{0.9-1}{0.25} \le \frac{x-1}{0.25} \le \frac{1.54-1}{0.25}\right]$$



i.e, $P[-0.4 \le z \le 2.16] = P(-0.4 \le z \le 0) + P(0 \le z \le 2.16)$

$= P(0 \le z \le 0.4) + P(0 \le z \le 2.16)$

$= 0.1554 + 0.4846$

$= 0.64$.

2. X is a normal variate with mean 30 and S.D 5. Find the probabilities that (i) $26 \le X \le 40$ (ii) $X \ge 45$ (iii) $|X-30| > 5$.

Solution: Given $\mu = 30; \sigma = 5$.

We know that $z = \frac{X-\mu}{\sigma} = \frac{X-30}{5}$

(i) $P(26 \le X \le 40) = P\left(\frac{26-30}{5} \le \frac{X-\mu}{\sigma} \le \frac{40-30}{5}\right)$

$= P(-0.8 \le z \le 2)$

$= P(-0.8 \le z \le 0) + P(0 \le z \le 2)$

$= P(0 \le z \le 0.8) + P(0 \le z \le 2)$

$= 0.2881 + 0.4772$

$= 0.7653.$

(ii) $P(X \ge 45) = P\left(\frac{X-\mu}{\sigma} \ge \frac{45-30}{5}\right) = P(z \ge 3)$

$= 0.5 - P(0 \le z \le 3)$

$= 0.5 - 0.4987 = 0.0013.$

(iii) $P(|X-30| \le 5) = P(-5 \le X-30 \le 5)$

$= P(25 \le X \le 35)$

$= P\left(\frac{25-30}{5} \le \frac{X-30}{5} \le \frac{35-30}{5}\right)$

$= P(-1 \le z \le 1) = 2P(0 \le z \le 1)$

$= 2 \times 0.3413 = 0.6826.$

$\therefore P(|X-30| < 5) = 1 - P(|X-30| \le 5)$

$= 1 - 0.6826$

$= 0.3174.$

3. The savings bank account of a customer showed an average balance of Rs. 150 and a standard deviation of Rs. 50. Assume that the account balances are normally distributed.

(i) What percentage of account is over Rs 200 ?

(ii) What percentage of account is between Rs. 120 and Rs. 170 ?

(iii) What percentage of account is less than Rs. 75. ?

Solution: Given $\mu = 150$; $\sigma = 50$.

(i) $P(X \geq 200) = P\left(\frac{X-150}{\sigma} \geq \frac{200-150}{50}\right) = P(z \geq 1)$

$= 0.5 - P(0 \leq z \leq 1) = 0.5 - 0.3413$

$= 0.1587$.

$\therefore$ 15.87 % of account is over Rs. 200.

(ii) $P(120 < x < 170) = P\left(\frac{120-150}{50} < z < \frac{170-150}{50}\right)$

$= P(-0.6 < z < 0.4)$

$= P(0 < z < 0.6) + P(0 < z < 0.4)$

$= 0.2257 + 0.1554$

$= 0.3811$

$\therefore$ 38.11 % of account is between Rs 120 and Rs 170.

(iii) $P(X < 75) = P\left(\frac{X-150}{50} < \frac{75-150}{50}\right) = P(z < -1.5)$

$= 0.5 - P(0 < z < 1.5)$

$= 0.5 - 0.4332$

$= 0.0668$

$\therefore$ 6.68 % of account is less than Rs 75.

4. Suppose the heights of men of a certain country are normally distributed with average 68 inches and S.D = 2.5.
Find the percentage of men who are
(i) between $a = 66$ and $b = 71$ inches in height.
(ii) approximately 6 feet tall.

Solution: Given $\mu = 68$; S.D $\sigma = 2.5$

(i) $P(66 \le x \le 71) = P\left(\frac{66-68}{2.5} \le z \le \frac{71-68}{2.5}\right)$

$= P(-0.8 \le z \le 1.20)$

$= P(0 \le z \le 0.8) + P(0 \le z \le 1.2)$

$= 0.2881 + 0.3849 = 0.6730.$

$\therefore$ Approximately 67.3% men are between 66 and 71 inches in height.

(ii) Assuming heights are rounded off to the nearest inch.
Then we get $a = 71.5$ and $b = 72.5$.

$P(71.5 \le X \le 72.5) = P\left(\frac{71.5-68}{2.5} \le \frac{X-68}{2.5} \le \frac{72.5-68}{2.5}\right)$

$= P(1.4 \le z \le 1.8)$

$= P(0 \le z \le 1.8) - P(0 \le z \le 1.4)$

$= 0.4641 - 0.4192 = 0.0449.$

Approximately 4.5% of the men are 6 feet tall.

5. The life time of an electric component is normally distributed with mean value of 250 hours and standard deviation of σ hours. Find the values of σ so that the probability of the component to have life between 200 and 300 hours is 0.7.

Solution: Given $\mu = 250$ and $P(200 \le X \le 300) = 0.7$.

Now, $P(200 \le X \le 300) = P\left(\frac{200-250}{\sigma} \le \frac{X-250}{\sigma} \le \frac{300-250}{\sigma}\right)$

$= P\left(\frac{-50}{\sigma} \le z \le \frac{50}{\sigma}\right) = 2P\left(0 \le z \le \frac{50}{\sigma}\right) = 0.7$ (given)

i.e, $P\left(0 \le z \le \frac{50}{\sigma}\right) = 0.35$. From normal table,

$\frac{50}{\sigma} = 1.04 \Rightarrow \sigma = \frac{50}{1.04} = 48.077.$

6. If M.G.F $= M_x(t) = e^{3t+8t^2}$ find $P(-1 < x < 9)$.

Solution:

For $X \sim N(\mu, \sigma^2)$, $M_x(t) = e^{\mu t + \frac{\sigma^2 t^2}{2}}$

Given $M_x(t) = e^{3t+8t^2}$.

Comparing we get $\mu = 3$, $\frac{1}{2}\sigma^2 = 8$, $\therefore \sigma^2 = 16$ i.e $\sigma = 4$.

To find $P(-1 < x < 9) = P\left[\frac{-1-3}{4} < \frac{x-3}{4} < \frac{9-3}{4}\right]$

$= P\left[-1 < z < \frac{3}{2}\right] = P[0 < z < 1] + P\left[0 < z < \frac{3}{2}\right]$

$= 0.3413 + 0.4332 = 0.7745$

Exercise:

1. A certain type of storage battery lasts on the average 3.0 years with S.D of 0.5 years. Assuming that the battery lives are normally distributed, find the probability that a given battery will last less than 2.3 years. Ans: $P(X < 2.3) = 0.0808$.

2. An electrical firm manufactures light bulbs that have a length of life which is normally distributed with $\mu = 800$ hrs. and $\sigma = 40$ hrs. Find the probability that a bulb burns between 778 and 834 hours. Ans: $P(778 < X < 834) = 0.5111$.

3. The mean yield of one acre plots is 662 kgs with S.D 32. Assuming normal distribution, how many one acre plots in a batch of 1000 plots would you expect to yield.
   (i) over 700 kgs (ii) below 650 kgs
   (iii) What is the lowest yield of the best 100 plots?
   Ans (i) 117 plots
   (ii) 352 plots
   (iii) 702.96 kgs

4. In an examination the marks obtained by the students in Maths, Physics and Chemistry are normally distributed about mean 50, 52, 48 and S.D 15, 12, 16 resply. Find the probability of securing a total mark of (i) 180 or above (ii) 90 or below. Ans: (i) $P(T \geq 180) = 0.1151$ (ii) $P(T \leq 90) = 0.0082$.

5. If x is normally distributed with $\mu = 12$ and $\sigma = 4$. Find the probability of (i) $X \geq 20$ (ii) $X \leq 20$ (iii) $0 \leq X \leq 12$.
   (iv) Find a when $(X > a) = 0.24$
   (v) Find b and c when $P(b < X < c) = 0.5$ and $P(X > c) = 0.25$.
   Ans: (ii) 0.9772 (iii) 0.4987 (iv) $a = 14.84$ (v) $b = 9.32$, $c = 14.68$.
   (i) 0.0228

DISCUSSION QUESTIONS

Try to answer the following questions:

1. What is normal distribution?

2. How do you determine normal distribution?

3. Why is normal distribution used?